Fabric IQ in the Real World

Chris Hyde

Matt Gordon

Thank You To Our Sponsors
For Making SQL Saturday Possible!

KEEP YOUR TAX DOLLARS LOCAL!

The PEACH Education Tax Credit lets you redirect your Georgia state income tax dollars directly to Innovation Academy — at no additional cost to you. This is a 100% dollar-for-dollar tax credit (not a deduction).

WHY THIS MATTERS

Instead of sending your tax dollars to the state's general fund, you can directly support:

- Mentorships & internships
- STEM and innovation programs
- Curriculum development
- Teacher professional growth
- Real-world student experiences

ELIGIBILITY LIMITS

- **Single Filer:** UP TO $2,500
- **Filing Jointly:** Up To $5,000
- **LLC Member or Pass Through Entity**: Up to $25,000
- **C or S Corporation:** up to 75% of annual Georgia income tax liability

Same Taxes. Greater Impact.

STEP 1: APPLY

- Apply for the PEACH Tax Credit
- Select "FCS Innovation Academy"
- Receive approval within approximately 10 days

STEP 2: DONATE

- Follow the instructions in your approval letter
- Submit your donation (online or by mail)
- Must be completed within 60 days of approval

STEP 3: CLAIM

- File Form IT-QED-TP2 with your Georgia tax return
- Receive your full dollar-for-dollar state tax credit

gfpe.org/tax_credit/

A Very Special Thank You To Our Community Partner Innovation Academy!

Please Support Via Tax Credit Eligible Donation That Goes DIRECTLY To Innovation Academy

Lunch Hour Sponsor Sessions

Microsoft SQL and AMD in the Era of AI
by Bob Ward

DBA Survival Kit: Redgate Monitor + Toolbelt
by Chris Reyes

Stop Firefighting: Build a Semantic Layer Ready for AI
by Anouk Gorris

Sponsor Session Attendees Can Win A $50 Amazon Gift Card Each
Two Raffle Gift Cards Per Session

Session Evaluations

SQL Saturday Atlanta, AI & BI - Session Evaluation

SQL Saturday Atlanta Closing Ceremony
5PM In The Auditorium
Must Be Present To Win Raffle Prizes

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About Me

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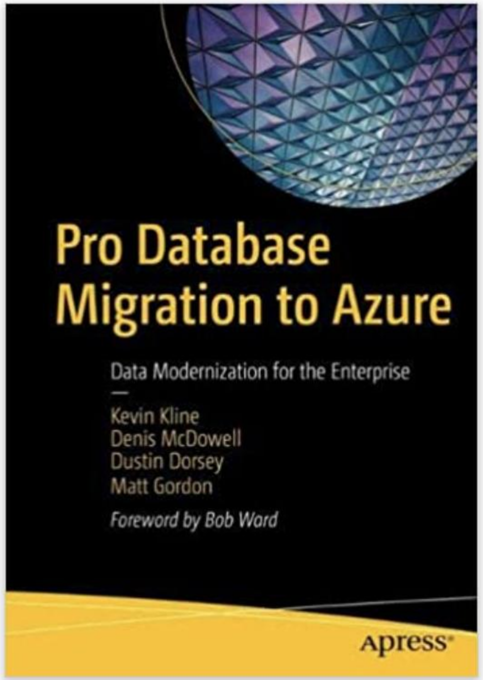

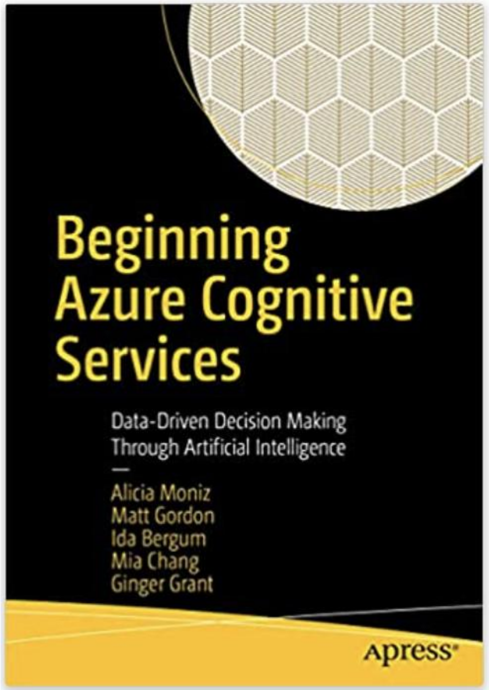

We Want Your Feedback

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What Era Are We In Now?

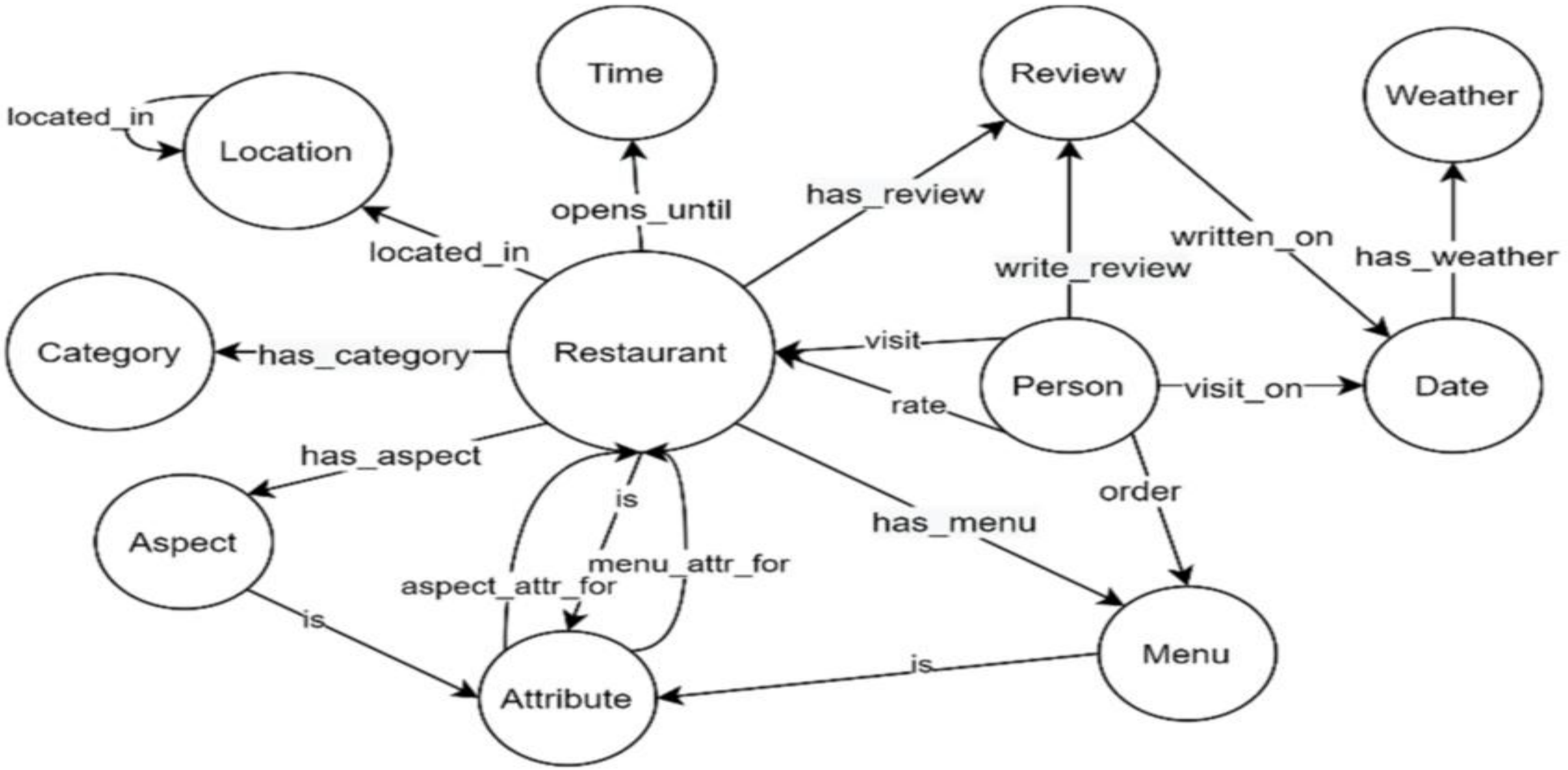
We're Nothing If Not Trendy…

- "At least 80% of governments will deploy AI agents to automate routine decision-making, enhancing efficiency and service delivery by 2028." – *Gartner*

- "Rapid AI agent adoption will overwhelm data and analytics operations built by and for humans." – *Gartner*
  - "Boooooo" – *humans who work in data & analytics*

Ontology explained

Applied ontology in information systems



Types of Agents in Microsoft Fabric

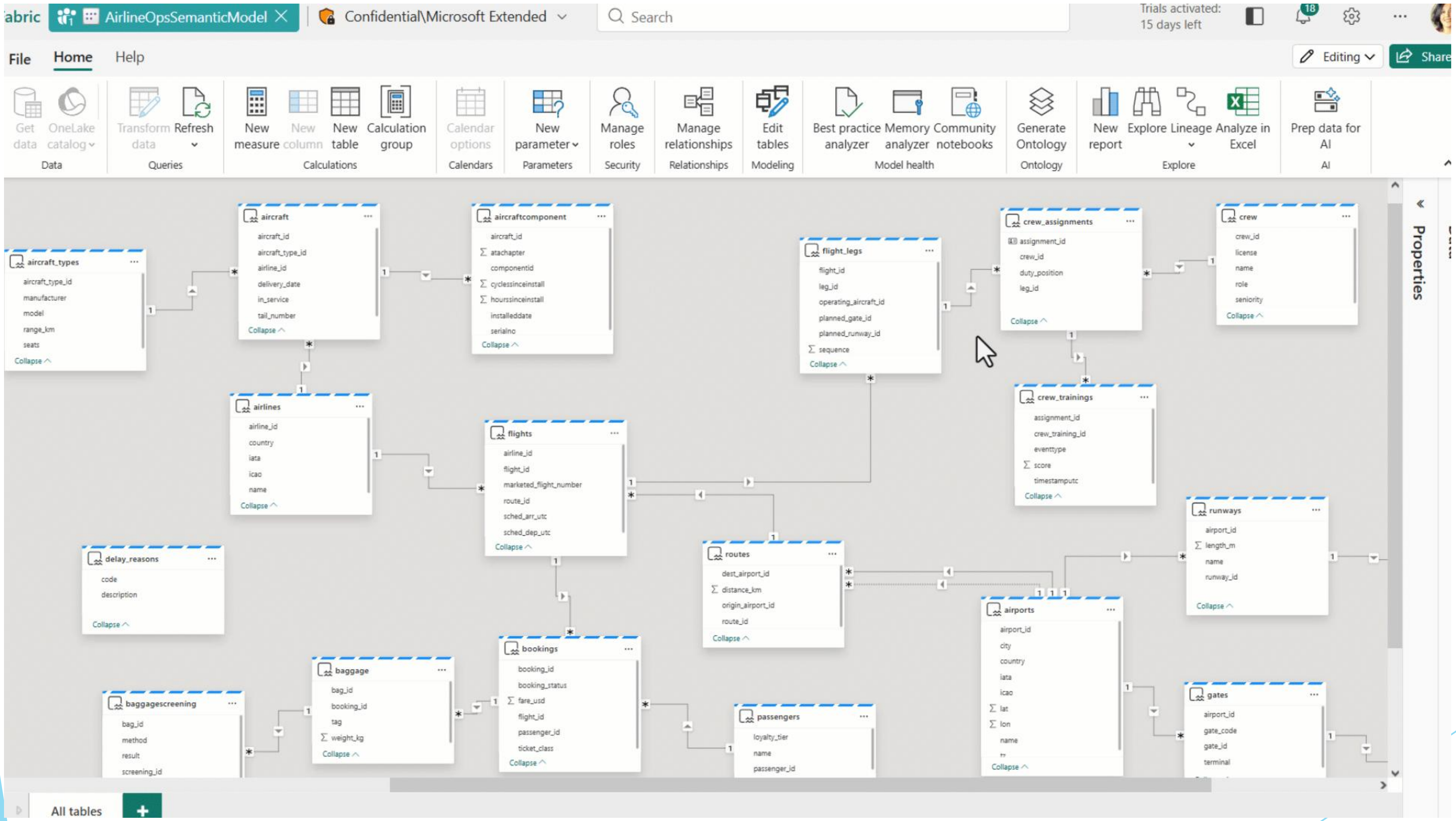
Data Agent

- You ask a question
- It translates your question into a query and gives you an answer
- Isolated investigations
- It takes no actions unless told

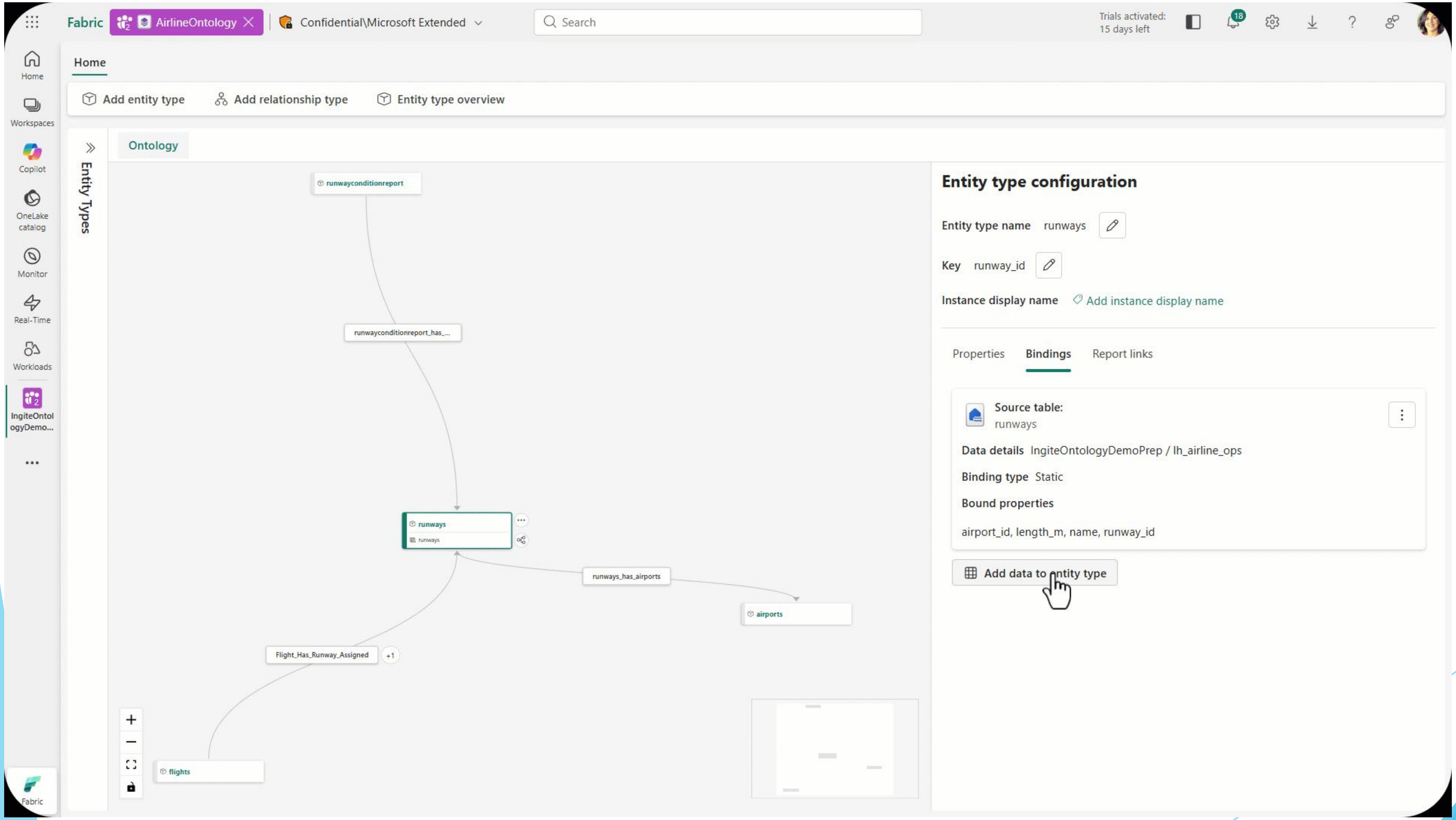
Operations Agent

- Can run constantly in the background
- Detects events and conditions
- It “reasons” and recommends a response
- It can take actions if permitted

Automated Ontology Generation



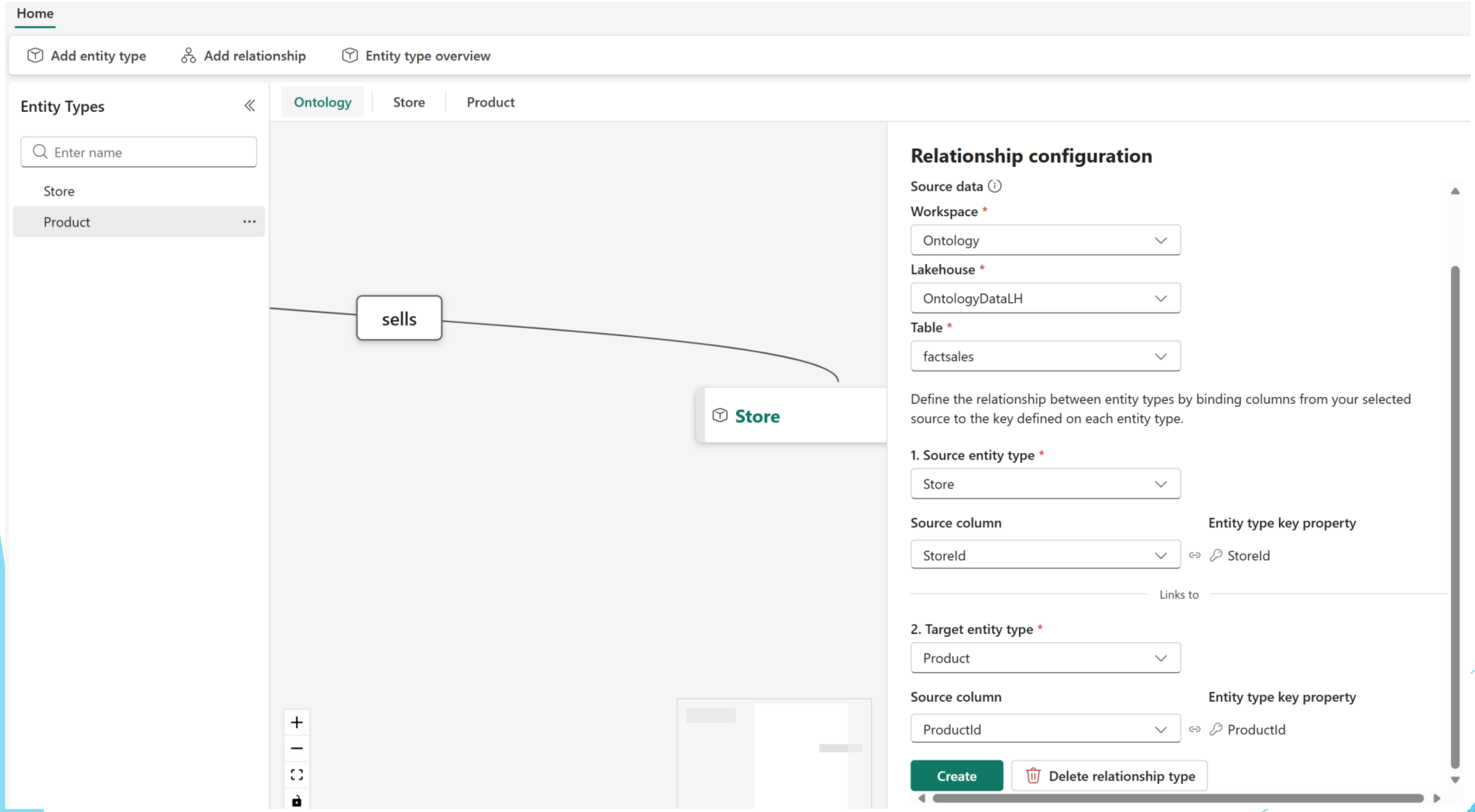
Sample Ontology with Enterprise Data



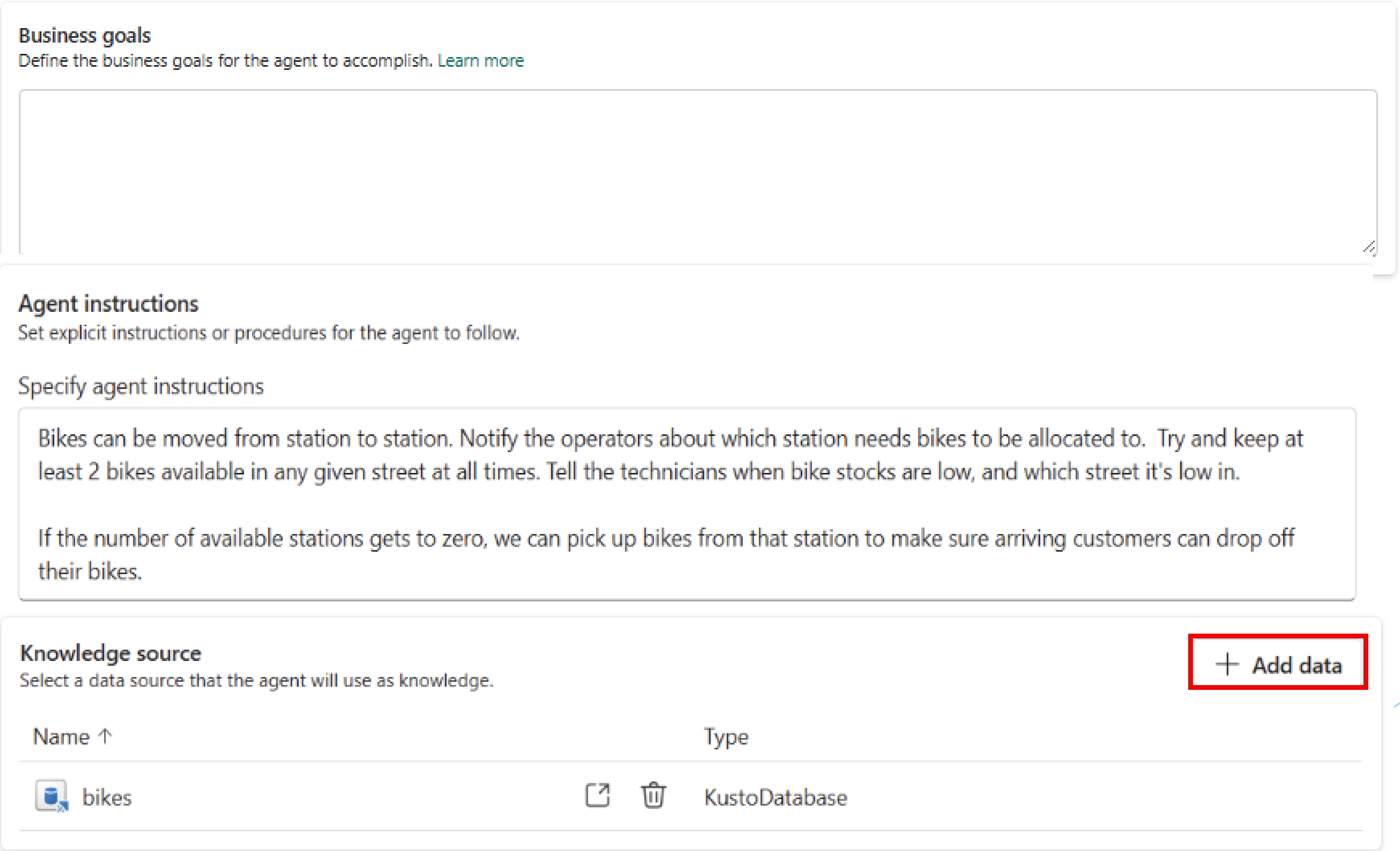
Ontology Concepts/Examples

- Entity type
  - “An abstract representation of a business object. It defines the logical model of an item.”
  - Vehicle, sensor, etc.
- Relationship type
  - “A definition that specifies how two entity types are connected.”
  - Located_at, monitoring_by, etc.
- Relationship instance
  - “A specific occurrence of a relationship type between two entity instances.”
  - Many sensors -> one vehicle

Relationship Configuration Example



How Do We Define An Operations Agent?



Who Sets All This Up and Maintains It?

- Who sets up the ontology?
- Who defines it?
- Who sets up the operations agents?
- Who is allowed to modify them?

Do I Need an Ontology and a Gold Layer?

- Ontologies and gold layers are both curated
- Business processes have approved their definition and contents
- Would the same users consume both?
- Will they be consumed the same way?

How Is Ontology Relevant?

- What is the relevance for human “agents” vs. AI agents?
- Is ontology more or less relevant for SMB vs. large enterprises?
- Is defining an ontology a relevant requirement for implementing Fabric?

Who's In Charge Of My Ontology and What Will It Charge Me?

- Who should in charge of establishing and governing our ontology?
- Who should be in charge of updating our ontology?
- I only have smaller capacities (F2, F4, F8, etc.) – will this eat up some to all of my CUs?

Is Fabric IQ Integral to Fabric?

- Should we be working with Fabric IQ now or wait for it to go GA?
- Will implementing Fabric IQ benefit every user that consumes data from Fabric?
- Is Fabric IQ ready for prime time?

Guidelines and Guardrails

- Reports of high capacity usage
- Very early days for this
- Segregate this capacity from anything else

Resources

- https://github.com/drjekyll325/presentations

- https://blog.fabric.microsoft.com/en-US/blog/trusted-ai-starts-with-microsoft-fabric-unified-real-time-intelligence-and-iq-context/
- https://learn.microsoft.com/en-us/fabric/iq/ontology/overview
- https://learn.microsoft.com/en-us/fabric/iq/ontology/tutorial-0-introduction?pivots=semantic-model
- https://learn.microsoft.com/en-us/fabric/iq/ontology/resources-capacity-usage
- https://sandervandevelde.wordpress.com/2025/12/25/first-look-at-fabric-ontology/

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